

ROVER 3D – GUM DISEASE MODULE

Use Case Document

Version 1.0

Module: Periodontal Visualization Engine

Primary System: Rover 3D (Unity)

Primary Users: Dentist, Hygienist

1. PURPOSE

The Gum Disease Module converts periodontal charting data into a clinically accurate 3D visualization of:

- Structural attachment loss
- Bone recession
- Gum recession
- Active inflammation

This module transforms a 2D periodontal chart into a dynamic 3D structural heatmap inside Rover 3D.

2. SYSTEM OVERVIEW

Periodontal charting is a diagnostic process used to evaluate:

- The structural integrity of the tooth support system
- The severity of gum disease
- Whether disease is active

Each tooth is supported by:

- Alveolar bone (foundation surrounding the roots)
- Periodontal ligament (shock-absorbing suspension tissue)
- Gum tissue seal (protective soft-tissue barrier)

Gum disease represents:

- Breakdown of the gum seal
- Downward collapse (resorption) of supporting bone

The periodontal chart measures exactly where and how much that support has failed.

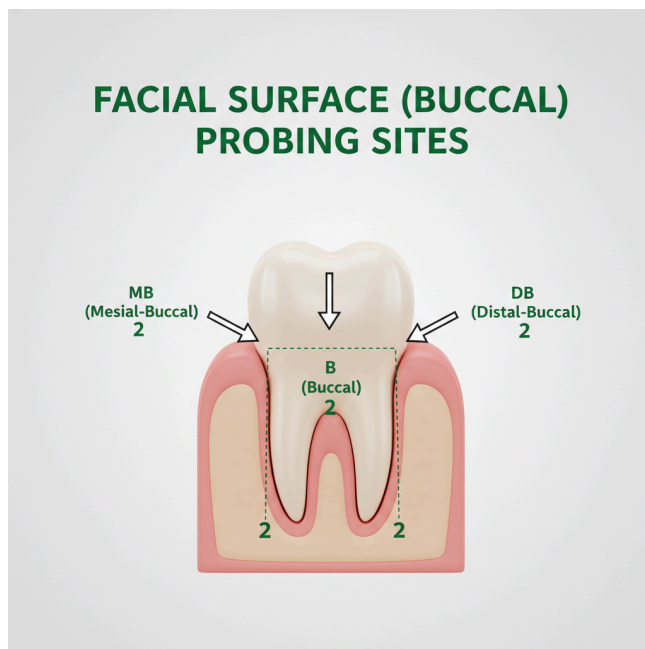
3. PRIMARY MEASUREMENTS

Each tooth contains 6 measurement sites.

3.1 Site Positions Per Tooth

Buccal (Facial) Surface

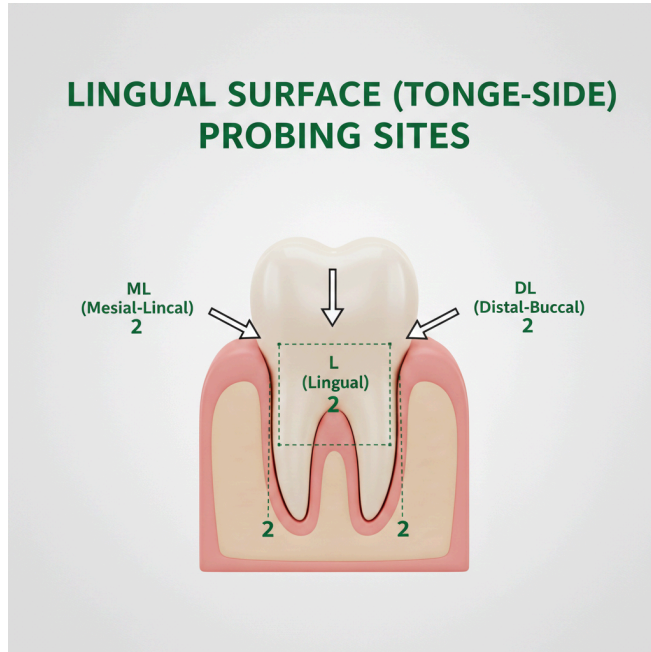
- MB – Mesial Buccal
- B – Mid Buccal
- DB – Distal Buccal



Lingual Surface

- ML – Mesial Lingual
- L – Mid Lingual

- DL – Distal Lingual



Per tooth:

- 6 probing depth values (millimeters)
- 6 bleeding indicators (boolean)

4. CLINICAL INTERPRETATION MODEL

4.1 Probing Depth (PD)

Definition: Distance (in millimeters) that a periodontal probe slides between gum and tooth.

Interpretation (Non-Table Format):

- A probing depth of 1–3 mm represents healthy attachment with intact supporting bone and gum seal.
- A probing depth of 4 mm represents early disease and the beginning of attachment breakdown.
- A probing depth of 5–6 mm represents moderate attachment loss and visible structural compromise.
- A probing depth of 7 mm or greater represents advanced attachment loss with significant structural collapse.

Probing depth indicates structural support loss. It reflects downward collapse of attachment and bone, not outward swelling.

Engineering analogy:

- Tooth = Structural pillar
 - Bone = Foundation
 - Probing depth = Crack depth along the pillar foundation
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4.2 Bleeding on Probing (BOP)

Definition:

Whether gum tissue bleeds when gently probed.

Interpretation:

- True → Active inflammation
- False → No active inflammation

Bleeding indicates infection activity but does not directly determine bone height.

5. DIAGNOSTIC LOGIC

Dentists evaluate severity using three dimensions:

Step 1 – Severity of Probing Depth

- Mostly 1–3mm → Healthy or Gingivitis
- Several 4mm → Early Periodontitis
- 5–6mm sites → Moderate Periodontitis
- ≥7mm sites → Severe Periodontitis

Step 2 – Bleeding Distribution

- Many bleeding sites → Active disease
- Deep pockets without bleeding → Stable chronic condition

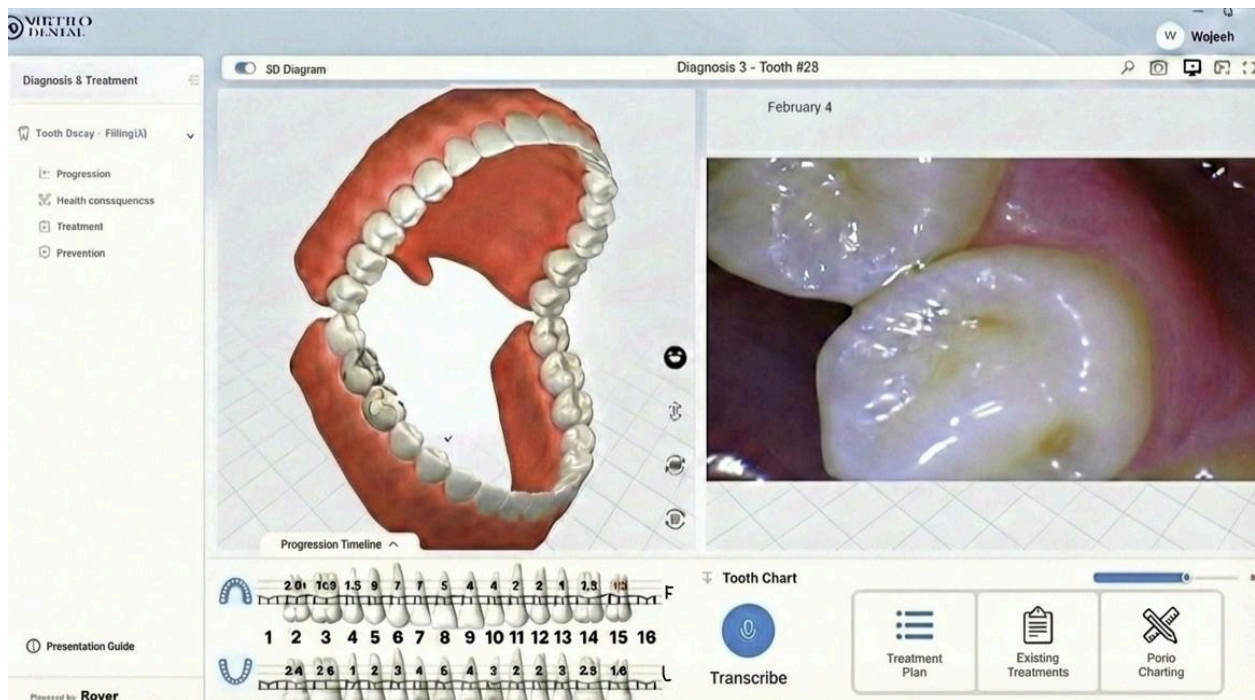
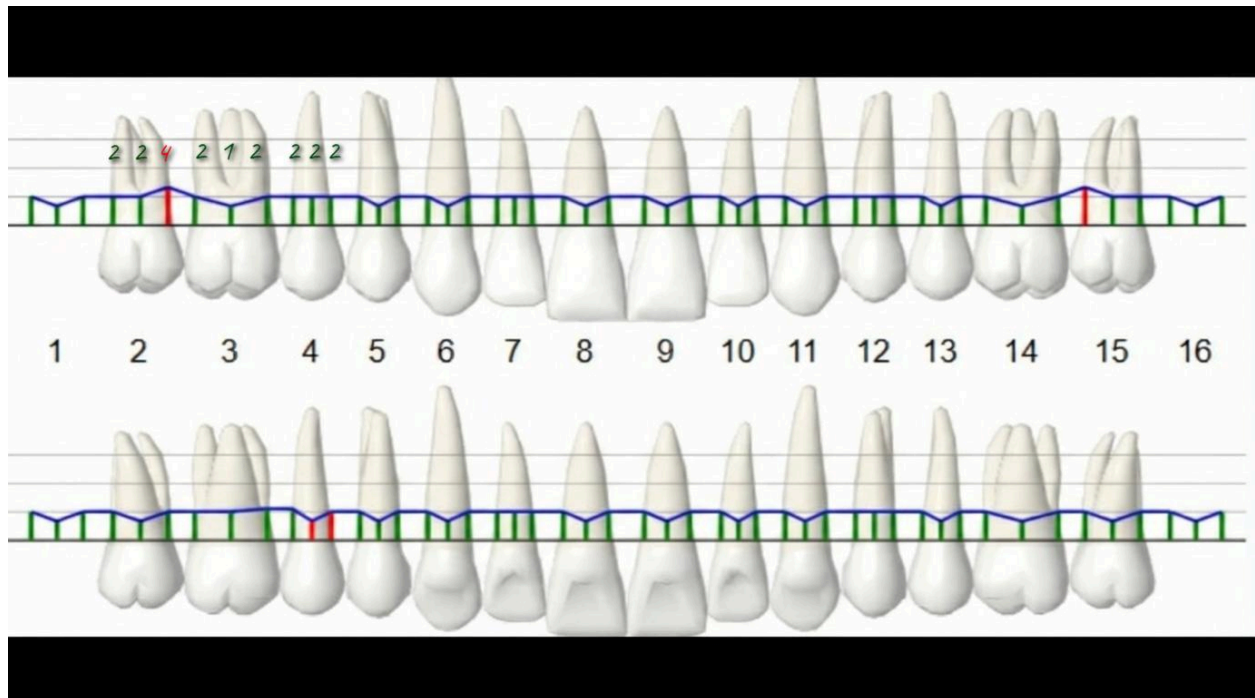
Step 3 – Pattern Recognition

- Localized (limited teeth)
- Generalized (most of the mouth)
- Isolated vertical defect (single deep crater)

These patterns inform treatment planning.

6. SYSTEM USE CASES

USE CASE 1 – Update UI with Perio Chart View



Actor: Dentist / User

Trigger: User selects "Perio Charting" button in the UI.

Preconditions:

- Tooth Chart view is currently displayed.

- Periodontal module is enabled.

Flow:

1. User clicks "Perio Charting".
2. The standard tooth chart in the lower left panel is replaced with the periodontal chart layout (as shown in the reference image).
3. Upper and lower arch selector buttons appear on the left side. The default arch displayed is the upper arch
4. When an arch button is selected:
 - The corresponding arch (Upper or Lower) is displayed.
 - The selected button is visually highlighted (active state).
5. On the right side of the chart:
 - The letter "F" represents the Facial (Buccal) surface.
 - The letter "L" represents the Lingual surface.
6. The chart displays:
 - Six probing depth number per tooth. Three numbers on the facial (F) surface (top row) and three numbers on the lingual (L) surface.
 - Vertical bar visualization beneath numbers. The larger the number, the longer the bar
 - Color-coded values (green for 1–3mm, red for ≥ 4 mm).

Postcondition:

UI correctly reflects periodontal charting mode and allows arch-specific selection.

USE CASE 2 – Periodontal Data Entry (Manual, PMS Import, and Audio-Ready Architecture)

F and B are used interchangeably. They represent the same surface

Actor: Dentist / Hygienist / System

Trigger:

- Manual numeric entry inside Perio Chart
- Future PMS import (Open Dental)
- Future audio transcript entry inside Unity

Preconditions:

- Tooth identifiers available (1–32)
- System knows which teeth are missing
- 6 probing depth fields per tooth available (F: MB, B, DB; L: ML, L, DL)

- Data model supports structured per-tooth storage
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A. CORE ARCHITECTURE REQUIREMENT (Future-Proofing)

All periodontal data entry methods (manual, PMS import, audio) must write to the SAME internal data structure:

PerToothPerioData

- toothNumber
- surface (B/F or L)
- siteIndex (0–2 per surface)
- probingDepth (int)
- bleedingFlag (bool)

No UI logic should contain business rules that cannot be reused.

All entry paths must call a shared function:

ApplyPerioValue(toothNumber, surface, siteIndex, value)

This prevents rework when Open Dental import and audio entry are added.

B. MANUAL DATA ENTRY WORKFLOW

1. User selects a tooth in the periodontal chart.
 2. Cursor automatically appears above the first vertical line on the Facial (F) surface.
 3. Cursor movement logic:
 - Facial surface: moves LEFT → RIGHT across MB, B, DB
 - Lingual surface: moves RIGHT → LEFT across ML, L, DL
 4. User types a number (1–9).
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C. VALUE ENTRY BEHAVIOR

If user enters 1, 2, or 3:

- The number appears above the vertical line.
- The vertical bar remains at baseline height.
- The number and line remain default color (green).

If user enters 4 or above:

- The number appears above the vertical line.
- The vertical bar elongates proportionally.
- Both number and vertical line change color to red.

Example:

- Tooth 2 selected.
- User types "3" → number 3 appears, no height change.
- Cursor auto-advances.
- User types "5" → bar elongates, number & bar turn red.
- Cursor auto-advances.
- User types "1" → baseline height, green.

After entry:

- Cursor automatically moves to the next site.
- No manual click required.

D. ARCH TRANSITION LOGIC

Entry Order (Top Arch First Scenario):

1. Start at Tooth 1 Buccal (MB → DB, left to right).
2. Continue buccal measurements sequentially across the entire upper arch from Tooth 1 → Tooth 16.
3. After Tooth 16 Buccal is completed, reverse direction and begin Lingual on the same arch.
4. Lingual measurements proceed from Tooth 16 → Tooth 1 (reverse order across the arch).
5. After Tooth 1 Lingual is completed:
 - System auto-selects Lower Arch.
 - Entry begins at Tooth 32 Buccal.
6. Lower arch buccal measurements proceed from Tooth 32 → Tooth 17.
7. After Tooth 17 Buccal is completed, reverse direction and begin Lingual on the lower arch.
8. Lower arch lingual measurements proceed from Tooth 17 → Tooth 32.

If entry begins on the lower arch, the inverse sequencing applies:

- Start Buccal 32 → 17
- Then Lingual 17 → 32
- Then move to upper arch Buccal 1 → 16
- Then Lingual 16 → 1

Missing teeth must be skipped automatically.

The cursor move automatically in the order outlined above. the cursor should never stop on a missing tooth.

E. PMS IMPORT (OPEN DENTAL – FUTURE)

When data is imported:

1. PMS sends structured probing depths per tooth.
2. System maps PMS tooth numbering to Rover numbering.
3. System calls ApplyPerioValue() per site.
4. UI updates identically to manual entry.

No separate rendering logic permitted.

F. AUDIO ENTRY (FUTURE)

When hygienist speaks values (e.g., "Tooth 2: 3-5-1"):

1. Transcript parser extracts numbers.
2. Parser determines current tooth context.
3. Parser calls ApplyPerioValue() sequentially.
4. UI advances cursor automatically just as manual entry does.

Audio must not require alternate rendering logic.

Postcondition:

- Periodontal values stored in unified structure
- UI reflects visual rules (baseline vs red bars)
- Cursor traversal logic consistent
- System ready for structural rendering

Data entry system must remain extensible without duplication of business rules.

USE CASE 3 – Visualize Structural Bone Loss

Actor: Dentist

Trigger: Rendering cycle after data load.

Flow:

1. System reads probing depths.
2. Depth values normalized to structural damage values.
3. Each site drives its corresponding bone-loss blendshape.
4. Bone mesh lowers locally.
5. Gum mesh follows bone recession, but not in a 1:1 ratio. Clinically, soft tissue (gum) recession does not immediately mirror bone loss. In early disease (≤ 4 mm probing depth), there is typically little to no visible gum recession because inflammation can cause swelling that masks structural loss. Visible recession commonly begins around ≥ 5 mm probing depths. For modeling purposes, use the following simplified ratios:
 - ≤ 4 mm: 0mm visible gum recession
 - 5mm: begin mild recession (~ 0.25 mm per 1mm beyond 4mm)
 - 6–7mm: moderate recession (~ 0.6 mm recession per 1mm beyond 4mm)
 - ≥ 8 mm: advanced recession (~ 0.7 – 0.8 mm recession per 1mm beyond 4mm)

This allows bone to drop first, with gum gradually exposing root structure thereafter, creating biologically realistic progression.

- Adjust the gum blend shapes accordingly
- 6. Root exposure increases proportionally.

Postcondition:

3D arch visually matches attachment loss pattern.

USE CASE 4 – Visualize Inflammation

If this is going to require significant artwork adjustment (more than 5 hours), move this to the next release, not MVP.

Actor: Dentist

Trigger: Bleeding flags processed.

Flow:

1. System evaluates bleeding boolean per site.
2. If true:
 - Increase gum redness

- Apply subtle swelling shader
- Increase surface gloss
- 3. If false:
 - Maintain baseline gum material

Postcondition:

Inflammation appears visually distinct from structural damage.

USE CASE 5 – Bone Profile Representation

The 2D periodontal chart displays a wave-like bone profile.

Unity must represent this by:

- Creating localized vertical bone dips per site
- Generating crater-like defects between teeth when adjacent sites are deep
- Preserving circumferential asymmetry

This creates anatomically accurate structural realism.

USE CASE 6 – Disease Progression Simulation

Actor: Dentist presenting to patient

Trigger: Adjustment of progression slider.

Flow:

1. Slider multiplies normalized structural damage values.
2. Bone lowers gradually over time.
3. Gum recedes proportionally.
4. Inflammation visuals scale with severity when bleeding present.

Postcondition:

Patient observes progressive collapse of support.

USE CASE 7 – Enter Periodontal Treatment

This must be done in a way that allows future treatment entry channels like audio and open dental import without any rework

Clinical Context Card Tooth #1

Fillings

Crown / Bridge

Endo

Perio

Oral

SRP (4+ Teeth) SRP (1-3 Teeth) Scaling (Full Mouth)

Quadrant Selector

UR UL

LR LL

Cancel Save

Actor: Dentist

Trigger: User double-clicks on a tooth within the Perio Chart.

Preconditions:

- Perio Chart view active.
- Tooth exists (not missing).

Flow:

1. User double-clicks a tooth in the perio chart.
2. Clinical Context Card appears.
3. On the left navigation of the Context Card, user selects "Perio".
4. The Perio Treatment entry interface is displayed.

The Perio Treatment interface contains three primary treatment options presented as tabs:

TAB 1 – SRP (4 or More Teeth per Quadrant) [Default]

This is the default tab when the Perio panel opens.

Displayed Elements:

- Quadrant selectors:
 - UR (Upper Right)
 - UL (Upper Left)
 - LR (Lower Right)
 - LL (Lower Left)
- User selects one or more quadrants.

This represents:

Scaling and Root Planing (SRP) on four or more teeth per quadrant.

Upon selection:

- Selected quadrant(s) are visually highlighted.
 - Treatment is associated with that quadrant.
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TAB 2 – SRP (1–3 Teeth)

This tab is used when SRP is performed on fewer than four teeth within a quadrant.

Displayed Elements:

- Primitive mini-tooth chart inside the Context Card.
- User can select individual teeth (1–3 teeth maximum per quadrant selection rule).

Upon selection:

- Selected teeth are visually highlighted.
 - Treatment is assigned at the tooth level.
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TAB 3 – Scaling in the Presence of Mild to Moderate Gingival Inflammation

This tab represents:

Scaling in the presence of generalized mild to moderate gingival inflammation.

Displayed Elements:

- No tooth-level selection required.
- Single action selection confirms treatment entry.

Postcondition:

- Selected treatment type stored in structured treatment object.
- Treatment associated to either quadrant-level or tooth-level based on selection.
- UI reflects applied treatment.

Future Architecture Requirement:

Treatment entry must be stored in structured format independent of UI (similar to periodontal data entry), to allow:

- PMS export
- Billing code mapping
- Analytics
- Automated treatment plan generation

TreatmentData object should contain:

- treatmentType (SRP_4plus, SRP_1to3, GingivalScaling)
- scopeType (Quadrant or Tooth)
- affectedQuadrants[]
- affectedTeeth[]

The interface must remain extensible without rewriting logic when additional periodontal procedures are added.

7. FUNCTIONAL REQUIREMENTS

FR1 – Six independent regions per tooth.

FR2 – Separation of channels:

- Probing depth drives structure
- Bleeding drives inflammation

FR3 – Support asymmetry:

One surface may be severe while others remain healthy.

FR4 – Crown integrity preservation:

Crown enamel must not deform.

FR5 – Circumferential modeling:

Bone loss must wrap around root surfaces instead of acting as a uniform global reduction.

8. VISUAL RULE SUMMARY

Visual output rules are as follows:

- If probing depth is 1–3 mm, there is no structural deformation.
 - If probing depth is 4 mm, there is mild bone depression.
 - If probing depth is 5–6 mm, there is visible recession and increasing root exposure.
 - If probing depth is 7 mm or greater, severe crater-like collapse is shown.
 - If bleeding is present, the gum displays an inflamed red shader regardless of structural depth.
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9. EDGE CASE HANDLING

- Deep pocket without bleeding → Chronic stable disease visualization
 - Bleeding without pocket → Gingivitis (inflammation only)
 - Single severe site → Vertical defect representation
 - Multiple adjacent severe sites → Generalized structural collapse
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10. SUCCESS CRITERIA

The module is successful if:

- Dentists can visually identify worst sites without relying on numeric overlays
 - Buccal and lingual differences render correctly
 - Bone dips accurately match probing measurements
 - Crown structure remains untouched
 - Performance remains stable under full-arch load (32 teeth × 6 sites)
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11. FINAL SYSTEM DEFINITION

Periodontal charting is a six-point structural scan per tooth measuring:

- Structural attachment loss (probing depth)
- Active inflammation (bleeding)

Rover 3D must:

- Map 6 depth values per tooth to 6 independent structural blendshapes
- Map bleeding values to inflammation shaders
- Preserve per-tooth independence
- Support asymmetrical attachment loss
- Support full-arch disease pattern recognition

The Gum Disease Module functions as a 3D structural representation of periodontal attachment failure within the Rover ecosystem.

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